Theory Note 7 of 8

Windows Logon Types 2, 3, 4, 5, 7, 10 and 11

Network Traffic Analysis, Nmap and Windows Logging | Week 2

1. Overview

Every successful or failed logon (events 4624 and 4625) carries a logon type, a number that says how the user or service tried to get in. The same account logging on is normal as an interactive session at a desk and suspicious as a remote desktop session at 3 AM. Reading the logon type together with the source address and the time is one of the quickest ways to judge a logon event.

2. Logon types

Type	Name	Meaning	Normal example	Suspicious example
2	Interactive	The user logs on at the machine's own keyboard or console.	An employee signs in to their own laptop in the morning.	A local console logon at 3 AM on a server nobody is working on.
3	Network	Access to a resource over the network, such as a shared folder. Credentials are not left on the target.	A user opens a mapped drive on the file server.	Hundreds of type 3 logons from one workstation to many hosts, which suggests lateral movement or pass-the-hash.
4	Batch	A scheduled task or batch job runs under a user account.	A nightly backup job running under a service account.	A new scheduled task running as an administrator that nobody created.
5	Service	A Windows service starts using its configured account.	The SQL Server service starts at boot.	A newly installed service starting from an unusual path or under a user account (see event 7045).
7	Unlock	A user unlocks a workstation that was locked.	A user comes back from lunch and unlocks the PC.	An unlock on a machine while the owner is on leave or in another location.
10	RemoteInteractive	A Remote Desktop (RDP) or Terminal Services logon.	An administrator connects by RDP to a server from the management network.	An RDP logon from an external IP address, or many failed 4625 events followed by a successful type 10 (a brute-force success).
11	CachedInteractive	A logon using cached domain credentials when the domain controller cannot be reached.	A laptop user signs in while travelling and offline.	A cached logon on a desktop that is always on the network, or on a machine the user never uses.

3. What to check alongside the type

- Account Name, and whether it is a normal user, a service account or an administrator.
- Source Network Address and Workstation Name, to see where the logon came from.
- Logon Process and Authentication Package (NTLM or Kerberos).
- The time of day, compared with that user's normal working pattern.

Key takeaway

Types 3 and 10 from the internet or from unexpected internal hosts deserve the fastest review. A failed type 3 or 10 repeated many times, followed by a success, is the classic brute-force pattern.